

CrewAI Architecture Comparison Report

Sequential vs. Hierarchical vs. Single-Agent | Cost, Quality, Latency & Architecture Selection

Objective

This report compares three approaches for the same Notion competitor-research workflow: a Single-Agent LangGraph solution, a Sequential CrewAI crew, and a Hierarchical CrewAI crew. The comparison focuses on output quality, token usage, estimated monetary cost, latency, and the practical trade-offs of each architecture.

Overall Comparison

Architecture	Quality	Avg Total Tokens	Avg Est. Cost	Avg Latency
Single-Agent LangGraph	11.0/15	~3,615	~\$0.00076	~10.49 s
Sequential CrewAI	12.0/15	64,484.67	\$0.012420	18.45 s
Hierarchical CrewAI	9.3/15	61,431.00	\$0.012613	46.23 s

Key result: Sequential CrewAI achieved the highest manual quality score, while Single-Agent LangGraph was substantially more efficient in tokens, latency, and estimated cost. Hierarchical CrewAI had the highest latency and slightly higher estimated cost than Sequential despite using fewer total tokens.

Quality Comparison

Sequential CrewAI (12.0/15): This was the highest-quality configuration in the experiment. Its fixed research → analysis → marketing pipeline matched the structure of the problem well and kept responsibilities clearly separated.

Single-Agent LangGraph (11.0/15): The single agent produced slightly lower manual quality, but it remained competitive while requiring dramatically fewer model calls and tokens.

Hierarchical CrewAI (9.3/15): The manager/delegation layer did not improve quality for this workflow. The additional coordination introduced overhead without a corresponding quality benefit.

Cost & Efficiency Notes

The recorded three-run averages show a large efficiency difference between the single-agent and multi-agent approaches. Single-Agent LangGraph used approximately 3,615 total tokens per run, compared with 64,484.67 for Sequential CrewAI and 61,431 for Hierarchical CrewAI.

The estimated average costs were approximately \$0.00076 for Single-Agent, \$0.012420 for Sequential, and \$0.012613 for Hierarchical. Although Hierarchical used fewer total tokens than Sequential, it generated more completion tokens, which resulted in a slightly higher estimated cost. Therefore, total tokens alone should not be treated as the complete measure of monetary cost.

The multi-agent approaches consumed approximately 17–18× more tokens than the Single-Agent baseline. This additional usage must be justified by meaningful gains in specialization, quality, delegation, or reliability.

Sequential vs. Hierarchical

Architecture	Pros	Cons	Best Used When
Sequential	Predictable order; clear dependencies; easy debugging; lower latency than hierarchical.	Less flexible; fixed workflow; limited dynamic delegation.	The task naturally follows a known sequence such as research → analysis → writing.
Hierarchical	Manager can delegate, coordinate, review, and adapt work dynamically.	Higher latency and coordination overhead; more model usage; can reduce reliability on simple workflows.	The task is complex, open-ended, or requires dynamic delegation and review.

Single-Agent vs. Multi-Agent

The experiment shows that specialization can provide a quality benefit when the business problem naturally separates into distinct stages. Sequential CrewAI improved the manual quality score from 11.0/15 to 12.0/15 compared with the Single-Agent baseline.

However, the quality improvement was relatively small compared with the increase in model usage. The Single-Agent approach remains the strongest option when cost, latency, simplicity, and resource efficiency are more important than the additional quality obtained through specialization.

Was the Multi-Agent Crew Worth It?

For this specific Notion competitor-research task, Sequential CrewAI was the better multi-agent architecture because it achieved the highest quality score while avoiding the much larger latency of the hierarchical manager. However, the overall multi-agent approach was not the most cost-efficient choice.

The Hierarchical configuration was not justified for this relatively well-defined workflow. Its 46.23-second average latency and 9.3/15 quality score indicate that the manager layer added coordination overhead without delivering a corresponding benefit.

Final Recommendation

Single-Agent LangGraph: Best when minimizing cost, latency, token usage, and implementation complexity is the primary objective.

Sequential CrewAI: Best multi-agent choice for this workflow because the work has a clear fixed order and specialization produced the highest quality.

Hierarchical CrewAI: Best reserved for substantially more complex workflows where dynamic delegation, manager review, and coordination provide enough value to justify the additional overhead.

Conclusion

There is no universally best agent architecture. The appropriate choice depends on the task's structure and the desired trade-off between quality, cost, latency, and complexity. For the Notion competitor-research workflow, Sequential CrewAI was the strongest multi-agent architecture, while Single-Agent LangGraph was the strongest efficiency-oriented architecture.

Note: Estimated costs are based on the token-rate assumptions used in the notebook and should be treated as experimental estimates rather than provider billing records.